

DroneArjuna (DA)

Priority Tasks Register

Derived from Plan Evaluation Report | ACSTL/DoTI/DAPE/DA/2026/001 | 10 August 2026

Document Reference	ACSTL/DoTI/DAPT/DA/2026/001
Source Document	DA_Plan_Evaluation_Aug2026.pdf — Plan Evaluation Report (07 Aug 2026)
Prepared By	Chandra Sekhar (Backend Lead), ACS Technologies Limited
Reviewed By	Director, Technology & Innovations
Date	10 August 2026
Classification	Restricted / Confidential
Status	Approved with Conditions — Actions A1 and A2 prerequisite for sign-off

1. Purpose

This document extracts and prioritises all actionable tasks from the DroneArjuna Plan Evaluation Report (ACSTL/DoTI/DAPE/DA/2026/001). It consolidates the ten Director-mandated Required Actions (A1–A10) and the fourteen product gaps (G1–G14) into a single ranked task register, with owners, deadlines, Definition of Done criteria, and milestone gates — the three structural elements found missing in the evaluated plan.

The evaluation rated the plan at ~72% maturity — **Approved with Conditions**. Actions A1 (Definition of Done per gap) and A2 (YOLOv8 baseline spike) are **prerequisites for Director sign-off** and must complete by 10 August 2026.

2. Priority Summary

All tasks are assigned one of four priority levels. The table below gives the count and earliest deadline for each level across the full task set (A1–A10 + G1–G14).

Priority	Count	Earliest Deadline	Rationale
CRITICAL	1 action (A2)	12 Aug 2026	YOLOv8 baseline — if mAP < 0.40 the entire ML pipeline estimate is wrong
HIGH	7 actions + 3 gaps	10 Aug 2026	Plan sign-off blockers and Window A foundation tasks; cascading dependencies
MEDIUM	3 actions + 5 gaps	21 Aug 2026	Scope / spec decisions that must precede dependent implementation tasks
LOW	6 gaps (execution)	31 Aug – 30 Sep	Well-understood implementation tasks within approved capacity model

3. Required Actions (A1–A10) — Director-Mandated

These ten actions address structural gaps in the *plan document itself*, not in the product. They were raised by the Director in the evaluation and must be resolved before or alongside Window A implementation work. Actions are sorted by priority then by deadline.

CRITICAL — Must complete before 12 August

Ref	Action	Owner	By	Definition of Done
A2	Establish YOLOv8 baseline on representative drone imagery Run pretrained YOLOv8n and YOLOv8s against 20–30 sample frames from drone-collected video. Record mAP50. If mAP50 < 0.40, the 8 PD estimate for G5 is materially wrong — revise G5 scope and estimates before 1 September.	Chandra Sekhar	12 Aug	mAP50 value recorded for both YOLOv8n and YOLOv8s on the sample set. If mAP50 < 0.40: revised G5 plan submitted to Director before Sep 1.

HIGH — Must complete by 10–13 August (pre-sign-off + Window A day 1–3)

Ref	Action	Owner	By	Definition of Done
A1	Define Definition of Done for every gap G1–G14 2–3 measurable pass/fail statements per gap. No milestone can be marked closed without an approved DoD. A half-day session with Chandra Sekhar + Indra is sufficient.	Chandra Sekhar + Indra	10 Aug	DoD document with 2–3 criteria per gap (G1–G14) shared with Director before sign-off. Each criterion is binary — pass or fail — not qualitative.
A8	Obtain current automated test coverage baseline Run pytest-cov against the full backend suite on Day 1. Record the line-coverage percentage. G10 targets >=80% but no baseline is documented in the plan.	Chandra Sekhar + Vaishnavi	10 Aug	pytest-cov report generated; current line-coverage % committed to the repo (e.g. in coverage/.coverage or a badge). Baseline number entered in the G10 tracking row.
A9	Add post-30-Sep acceptance gate criteria to the plan Define who reviews the deferred items list, what constitutes an acceptable deferral reason, and who signs off that the product is ready for the next phase (field trials / Indian Army demo / procurement documentation).	Director (review)	10 Aug	A single acceptance-gate paragraph added to the plan document, naming: (a) reviewer, (b) deferral criteria, (c) sign-off authority, (d) reference to procurement / demo context.
A4	Define job-state schema for G4 (PostgreSQL table vs RabbitMQ-native) Vaishnavi's G4 unit tests depend on a stable schema. This architectural decision must precede her Aug 10–14 work on the background worker.	Chandra Sekhar + Vaishnavi	10 Aug	Schema decision recorded (ADR or design note): PostgreSQL AnalysisJob table columns defined OR RabbitMQ-native state approach chosen and documented. Vaishnavi has reviewed and approved for her test scaffolding.

Ref	Action	Owner	By	Definition of Done
A3	Pin ffmpeg version; decide container packaging for G6 ffmpeg is not in the original tech stack. Must be added to requirements.txt with a pinned version, added to Dockerfile, and noted in the tech-stack amendment document (DA-DDD-001) that Vaishnavi is authoring in Window A.	Chandra Sekhar	Before G6 start (10 Sep)	ffmpeg-python (or av) at a pinned version present in requirements.txt. Dockerfile updated to install ffmpeg binary. DA-DDD-001 lists ffmpeg as an approved tech-stack addition.

MEDIUM — Complete before the relevant gap starts

Ref	Action	Owner	By	Definition of Done
A5	Add rollback / contingency procedure for each of the 6 milestones One sentence per milestone: (a) what is deferred if it fails, (b) who decides, (c) whether 30-Sep commitment is affected.	Chandra Sekhar	Before 21 Aug milestone	Contingency row added to each of the 6 milestone entries in the plan. Director has reviewed and accepted the deferral candidates.
A7	Specify polling interval for G9 Analyst frontend job-status updates 1s vs 5s polling has material UX and backend-load implications. Must be agreed before G9 implementation begins.	Indra + Chandra Sekhar	Before G9 start (7 Sep)	Polling interval agreed and documented in the G9 DoD. Backend rate-limit headroom verified for chosen interval at 10-drone load.
A6	Set WCAG standard explicitly to 2.1 AA for G12 accessibility pass Ambiguity on WCAG level creates scope risk. 2.1 AA is the correct baseline for a defence ground-control-station application.	Indra	Before G12 start (18 Sep)	G12 DoD (from A1) explicitly states WCAG 2.1 AA. 5-workspace audit checklist scoped to AA criteria only.
A10	Plan G13 load test against the original SRS NFR (sub-100ms C2 latency) Load test must validate the original SRS non-functional requirement, not merely confirm 10-drone survival. Specific latency target must appear in the G13 test plan.	Chandra Sekhar	Before Sep 18	G13 test plan updated with: target = <100ms C2 command round-trip at 10 simultaneous drones. Locust scenario covers telemetry streams + command dispatch concurrently. Pass/fail threshold stated in test script.

4. Gap Execution Order (G1–G14) — Prioritised by Dependency + Milestone

The 14 product gaps are sequenced by their dependency chain and milestone gates. Priority reflects blocking impact: a gap that blocks others or sits on the critical path to the 21-Aug milestone is rated higher than an independent late-window task.

Dependency chain: G4 (worker infra) → G5/G6 (ML inference + video extraction) → G7 (change detection) → G9 (analyst frontend). G1 (Cesium) → G2 (replay player). G3, G8, G10, G11, G12, G13, G14 are independent.

Ref	Gap	Track	P D	Priority	Milest one	Definition of Done
G8	MinIO object storage wiring Wire the deployed object storage service into the upload endpoint; confirm presigned URL flow.	Backend	4	HIGH	21 Aug	POST /api/analyst/upload returns a presigned URL. A test file uploaded via that URL appears in MinIO bucket.
G4	Analyst background worker (RabbitMQ) Celery worker infrastructure — broker config, worker/ directory, Docker Compose changes. Prerequisite for G5, G6, G7, G9.	Backend	6	HIGH	21 Aug	Celery worker starts without error. A test task enqueued via RabbitMQ is consumed and result persisted to DB. Vaishnavi schema review complete.
G1	3D visualisation scaffold (Cesium.js / terrain) Integrate Cesium.js; scaffold Map3DCanvas.tsx; terrain tiles loading in Fly workspace. Prerequisite for G2.	Frontend	10	HIGH	31 Aug	Cesium terrain tiles visible in Fly workspace on localhost. Mission waypoints rendered as 3D points on terrain. No console errors.
G11	CI pipeline (lint, type-check, test, build) GitHub Actions workflow running backend pytest + frontend Vitest + type-check + Docker build on every push.	Both	4	HIGH	21 Sep	GitHub Actions badge green on main branch. Pipeline runs in <10 min. A failing test causes the pipeline to fail.
G10	Automated test coverage >=80% Gap-fill tests for under-covered modules identified from A8 baseline. Ongoing across both windows.	Both	6	HIGH	30 Sep	pytest-cov report shows >=80% line coverage on the full backend suite. Coverage report committed to the repository.
G5	YOLOv8 object-detection inference Celery task running YOLOv8 inference on uploaded imagery. Depends on G4 + GPU/CUDA environment. Estimate contingent on A2 baseline spike (12 Aug).	Backend	8	MEDIUM	04 Sep	Inference endpoint returns bounding-box results + confidence scores for >=90% of a 30-frame reference set. >=1 detection result persisted to DB. Inference latency on reference hardware measured and recorded.
G6	Video-frame extraction pipeline (ffmpeg) Celery task extracting frames from uploaded video at configurable intervals. Depends on G4.	Backend	5	MEDIUM	14 Sep	Pipeline tested on 3 video formats (MP4, AVI, MOV). Extracted frames stored to MinIO. Frame count matches expected interval.

Ref	Gap	Track	P D	Priority	Milest one	Definition of Done
G3	Drone Inventory knowledge-base workflow Full InventoryWorkspace frontend tab + Elasticsearch deep integration replacing ILIKE queries.	Both	8	MEDIUM	30 Sep	InventoryWorkspace tab renders catalogue, spec detail, comparison, payload browser. Elasticsearch query returns results in <200ms for a 500-record corpus.
G9	G9 — Analyst frontend workspace Job submission form, detection-results viewer, status polling. Depends on G4 + G5.	Frontend	8	MEDIUM	14 Sep	Job launch flow complete end-to-end. Detection bounding-box overlay renders on test imagery. Job-status polling interval per A7 decision.
G7	Change-detection module Pixel-diff algorithm Celery task comparing before/after imagery. Depends on G5, G6.	Backend	6	LOW	14 Sep	Change-mask output stored to MinIO for 3 test image pairs. Change-detection map layer visible in Plan/Fly workspaces.
G2	Telemetry replay player (3D + timeline scrubber) Timeline scrubber replaying historical telemetry in the Cesium 3D view. Depends on G1.	Frontend	5	LOW	30 Sep	Replay scrubber plays back a recorded flight in the 3D view. Playback speed controls (1x / 2x / pause) functional.
G12	Accessibility pass (WCAG 2.1 AA across 5 workspaces) Focus rings, ARIA labels, colour contrast across Fleet, Fly, Monitor, Plan, Settings.	Frontend	5	LOW	30 Sep	All 5 workspaces pass axe-core automated check with 0 critical violations. Human visual review sign-off by Indra. WCAG 2.1 AA standard per A6.
G13	Load testing (10+ drone telemetry + API) Locust scenario validating sub-100ms C2 command latency at 10-drone concurrent load. Depends on G4–G8 stable.	Backend	4	LOW	21 Sep	Locust report shows <100ms C2 command round-trip at P95 under 10-drone concurrent telemetry + 3 concurrent mission uploads. Report committed to repo.
G14	Frontend automated test suite (zero > first pass) Vitest + React Testing Library first-pass coverage for critical frontend paths.	Frontend	6	LOW	30 Sep	>=1 Vitest suite file per workspace (5 total). At least the auth + mission creation flows covered. All passing on CI (G11).

5. Milestone Gate Table — Acceptance Criteria

Six milestone gates span the Aug–Sep 2026 delivery window. Each gate is now paired with the Director-suggested acceptance criterion from Section 5 of the evaluation report. A milestone may not be marked closed unless its acceptance criterion is verifiably met.

Date	Milestone	Gaps Closed	Acceptance Criterion	Contingency if Missed
21 Aug	Analyst job worker + MinIO end-to-end	G4 ✓, G8 ✓	Vaishnavi G4 schema review complete; presigned URL flow tested end-to-end.	If >3 days late: compress Window A by deferring G14 to Window B; notify Director same day.
31 Aug	3D visualisation scaffold usable; Window A close-out	G1 ✓, G14 partial	Cesium terrain tiles loading in Fly workspace; first Vitest suite passing on CI.	If G1 slips: notify Director; G2 automatically deferred to Window B. Window A closes on schedule regardless.
04 Sep	YOLOv8 inference producing real detections on test imagery	G5 partial	Confidence threshold defined; at least 10 labelled test frames validated. mAP50 recorded (from A2 baseline).	If G5 is under-performing: escalate immediately; G7 and G9 timelines compress. Draw G13 buffer PDs into G5 per plan Section 08.
14 Sep	Change-detection + video pipeline + Analyst UI end-to-end	G6 ✓, G7 ✓, G9 partial	ffmpeg pipeline tested on 3 video formats; UI job-launch flow complete end-to-end from submission to detection result display.	If G9 is behind: G12 (accessibility) and G3 frontend are the natural deferral candidates; Indra prioritises G9 completion.
21 Sep	CI pipeline green (backend + frontend); load-test report delivered	G11 ✓, G13 ✓	GitHub Actions passing on main branch; 10-drone load test meets sub-100ms C2 NFR per A10 test plan.	If G11 fails: coverage report still committed manually; G14 is the deferral candidate.
30 Sep	>=80% verified coverage; accessibility pass complete; all gaps closed or deferred	G10 ✓, G12 ✓, G14 ✓	Director sign-off on deferred items list; coverage report committed to repo. Half-day Director review scheduled 29 Sep to assess deferred items before sign-off.	30-Sep is the hard gate for Indian Army procurement readiness. Any unclosed gaps must have a formal deferral note signed by Director.

Watch items identified by the Director: (1) G5 YOLOv8 — most likely to require re-planning; the 12-Aug baseline spike is not optional. (2) Window B frontend load — ~23 PD against 22 PD capacity; G12 and G3 frontend are the natural cut candidates if Indra is behind at 14 Sep. (3) 30-Sep gate — half-day Director review must be booked for 29 Sep.

6. Capacity and Ownership Summary

Three team members are active on DroneArjuna for the Aug–Sep 2026 delivery period. The capacity model below is taken directly from the evaluated plan.

Team Member	Role	Window A Availability	Window B Availability	Assigned Gaps
Chandra Sekhar	Backend Lead	Full (100%)	Full (100%)	G4, G5, G6, G7, G8, G10 (backend), G11 (backend), G13 Actions: A1, A2, A3, A4, A5, A8, A10
Indra	Frontend Lead	Full (100%)	Full (100%)	G1, G2, G3 (frontend), G9, G11 (frontend), G12, G14 Actions: A1 (joint), A6, A7 (joint)
Vaishnavi	Full-stack / Design	25% (4 PD / 16 days) — reduced availability	Full (100%)	G3 (backend design), G4 (schema review), G10 (coverage) DA-DDD-001 design decisions addendum Actions: A4 (joint), A8 (joint)

Note on Vaishnavi's Window A constraint: The evaluation confirmed this capacity model is sound. Vaishnavi is assigned only tasks that run in parallel with Chandra Sekhar's work, avoiding context-switch overhead. At 25% availability, elapsed calendar time is approximately 4x the PD estimate — factor this into milestone tracking.

7. Immediate Actions — 10 August 2026 (Day One Checklist)

The following actions must be initiated or completed on the first day of Window A (10 August 2026). They are prerequisites for Director sign-off and unblock the critical-path work starting this week.

#	Action	Owner	Done When
1	Run pytest-cov — record current test coverage baseline (A8) Execute: <code>pytest --cov=app --cov-report=term-missing</code> Record the overall line % in the project tracking sheet.	Chandra Sekhar + Vaishnavi	Coverage % number recorded. If not 80% — escalate scope of G10 immediately.
2	Open DoD definition session (A1) Chandra Sekhar and Indra block 3 hours today to write 2–3 DoD statements for each of G1–G14. Use the G5 example from the evaluation as a template.	Chandra Sekhar + Indra	DoD document draft shared with Director by end of 10 Aug.
3	Book 12 August half-day for YOLOv8 baseline spike (A2) Reserve Chandra Sekhar's full morning on 12 Aug for the baseline evaluation. Obtain 20–30 representative drone-imagery frames before Aug 12.	Chandra Sekhar	Calendar blocked. Sample imagery frames available on the dev machine.
4	Schedule G4 schema design session for 13 August (A4) Chandra Sekhar + Vaishnavi agree on PostgreSQL AnalysisJob schema or RabbitMQ-native state approach. This must precede Vaishnavi's Window A work.	Chandra Sekhar + Vaishnavi	13 Aug session booked. Schema options prepared by Chandra Sekhar before the session.

#	Action	Owner	Done When
5	Add post-30-Sep acceptance gate criteria to plan (A9) Director adds a paragraph defining sign-off authority, deferral criteria, and connection to Indian Army procurement context.	Director review	Paragraph added to plan document before formal sign-off on 10 Aug.

Once A1 (DoD) and A2 (YOLOv8 baseline) are confirmed complete, the Director's conditional approval converts to full approval and Window A proceeds as planned. Next formal check-in: **21 August 2026 milestone review.**